

Carbon-Intelligent Workload Scheduling for AI Data Centers

A Deterministic Mathematical Optimization Approach

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Ethics Statement. This study was conducted in accordance with the principles of academic integrity and responsible research conduct. It draws solely on publicly available, aggregated electricity-grid data (hourly carbon intensity and carbon-free energy fractions from ElectricityMaps), used in compliance with the provider's terms of use; it involves no human participants and no personal or sensitive data, so research-ethics-committee approval was not required. All results are reported faithfully, without fabrication, falsification, or selective reporting, and every external source is cited. To support reproducibility, the deterministic model and the code and data used to generate every result are openly available in the accompanying public repository. Generative AI tools were used to assist with language editing, drafting, and the preparation of code and figures; the author verified all such output and takes full responsibility for the content of this thesis. The research aims to reduce data-center carbon emissions, and its limitations, notably the use of average rather than marginal emissions and the exclusion of embodied carbon and water, are stated explicitly so as not to overstate its benefits. The author declares no conflicts of interest.

Abstract

The electricity demand of AI workloads is growing faster than electricity grids can decarbonize, making the operational carbon of data-center computing a first-order sustainability concern. Because grid carbon intensity varies across both regions and hours, flexible workloads can be shifted in time and space toward cleaner electricity, yet existing carbon-aware schedulers either rely on heuristics with no optimality guarantee or treat the temporal and spatial dimensions separately. This thesis formalizes carbon-aware scheduling as a deterministic linear program that minimizes operational carbon by jointly choosing when and where flexible load runs across five structurally diverse grid regions, subject to seven operational constraints and driven by hourly carbon-intensity and carbon-free-energy signals from ElectricityMaps over a two-year horizon. The optimization model and its operational constraints are validated through a behavioural perturbation test and an integration check on real data. The study has three main findings. First, under unconstrained flexibility the model reduces carbon by up to 78 % relative to a no-optimization baseline. Second, a sensitivity analysis identifies geographic flexibility as the dominant driver of carbon savings. Third, an additive decomposition shows that routing load to cleaner regions accounts for almost all of the saving, while shifting load in time alone can increase carbon. The thesis contributes a joint spatio-temporal scheduling model that is provably optimal by construction and validated on real data, together with an empirical account of which form of flexibility drives carbon savings.

Keywords: carbon-aware scheduling, data centers, sustainable AI.

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1 Introduction

The rapid scaling of artificial intelligence has turned data-center electricity consumption into a material driver of global carbon emissions. Training and serving large models now account for a fast-growing share of compute demand, and the energy this requires has repeatedly been shown to carry a substantial carbon cost (Faiz et al., 2024; Patterson et al., 2022; Strubell, Ganesh, & McCallum, 2019). As AI workloads expand faster than electricity grids can decarbonize, the carbon emitted by the electricity a workload consumes while it runs, known as its operational carbon, has become a first-order sustainability concern and a recognized lever for aligning computing with climate-change mitigation (Kaack et al., 2022; Moore et al., 2025).

This carbon is partly controllable because the carbon intensity of grid electricity, the grams of CO₂-equivalent emitted per kilowatt-hour, is not constant. It varies hour to hour as the generation mix shifts between fossil and renewable sources, and it varies region to region with the local grid, so that a kilowatt-hour drawn from a hydro- and nuclear-rich grid at night can be many times cleaner than one drawn from a coal-heavy grid at peak. Because a large fraction of data-center load is delay-tolerant, an operator can reduce the carbon emitted to complete the same computation by deferring flexible work to cleaner hours and routing it to cleaner regions, without buying new hardware or curtailing the workload (Acun et al., 2023; Lindberg, Lesieutre, & Roald, 2022; Radovanovic et al., 2023).

Prior work, however, leaves two gaps. The first is methodological: much carbon-aware scheduling relies on heuristic rules that offer no guarantee of optimality and can violate operational limits, so an observed saving cannot be attributed to the approach rather than to one rule on one dataset (Radovanovic et al., 2023; Wijayawardana & Chien, 2025). The second concerns scope: the temporal and spatial dimensions are usually optimized separately rather than jointly, which leaves unresolved how much of the saving comes from when load runs versus where it runs (Liu et al., 2012; Sukprasert, Souza, Bashir, Irwin, & Shenoy, 2024). These gaps motivate the central research question of this thesis: *how can a deterministic optimization model, using carbon intensity and the carbon-free energy fraction as scheduling signals, minimize data-center carbon emissions through joint temporal and spatial workload scheduling across multiple grid regions, subject to realistic operational constraints?*

The thesis answers this question by formalizing carbon-aware scheduling as a linear program situated within a broader research framework that translates grid carbon signals into scheduling decisions (Colangelo et al., 2025). The model is driven by two hourly signals, carbon intensity and the carbon-free energy fraction, where the latter is the share of electricity supplied by carbon-free sources. Both signals are used because two hours with identical carbon intensity can differ in how structurally clean they are, and using both lets the model favour electricity that is clean because of renewable supply rather than electricity that is only incidentally low-carbon (Chien et al., 2023; Riepin & Brown, 2024). The model is solved exactly, optimizes both dimensions jointly, and is evaluated on a two-year backtest across five structurally diverse grid regions.

On this basis the thesis makes three main contributions. First, it develops a provably optimal joint spatio-temporal scheduling model that dispatches flexible load against real hourly grid carbon-intensity and carbon-free-energy signals, with the latter entering the objective as a structural-cleanliness discount; this distinguishes it both from heuristic schedulers and from the closest exact formulation, which optimizes dedicated clean-energy procurement rather than dispatch against grid signals. Second, it identifies geographic flexibility as the dominant determinant of achievable carbon savings, exceeding every other modelling parameter by an order of magnitude in a one-at-a-time sensitivity analysis. Third, it decomposes the realized saving into spatial and temporal components and shows that routing load to cleaner regions

accounts for almost all of it, whereas shifting load in time alone can increase carbon, thereby clarifying which form of flexibility actually drives the result.

The remainder of the thesis is organized as follows. Section 2 reviews the literature on carbon-aware computing and positions the model against prior heuristic and optimization-based work. Section 3 describes the carbon data, the five target regions, and the exploratory analysis that motivates the modelling choices. Section 4 presents the optimization model, its constraints, and its implementation. Section 5 reports the computational results, and Section 6 concludes with the study's limitations and directions for future work.

2 Literature Review

This section reviews the literature that situates the thesis and develops, in detail, the two gaps introduced earlier. It moves from the scale of the problem and the carbon signals that make scheduling possible, through the temporal and spatial flexibility the thesis exploits, to the contrast between heuristic and optimization-based methods, and closes by stating precisely how the present work is positioned.

2.1 The Carbon Footprint of Computing and AI

The motivation for carbon-aware scheduling begins with the scale of the problem. Data centers account for an estimated one percent of global electricity use, a figure that held stable through the 2010s as efficiency gains offset rising demand (Masanet, Shehabi, Lei, Smith, & Koomey, 2020). The rapid expansion of artificial intelligence is now raising it, with dedicated AI compute a fast-growing contributor whose projected demand is substantial within a few years (de Vries, 2023). Quantifying this footprint at the workload level has become an active research area, progressing from early measurements of training emissions (Strubell et al., 2019) to standardized accounting methods (Henderson et al., 2020; Lannelongue, Grealey, & Inouye, 2021) and end-to-end models for large language models (Faiz et al., 2024; Luccioni, Viguier, & Ligozat, 2023).

Across these accounting studies, a consistent theme emerges that motivates the rest of this thesis. Operational emissions depend not only on how much computation is performed but also on when and where the electricity is drawn, so that the same job run in a different region or at a different hour can emit several times more or less carbon (Dodge et al., 2022; Patterson et al., 2022). This locational and temporal sensitivity is what carbon-aware scheduling exploits. Operational carbon is not the only relevant dimension, however. A parallel literature stresses that the embodied emissions of hardware manufacturing (Gupta et al., 2021) and the water consumed for cooling and electricity generation (P. Li, Yang, Islam, & Ren, 2023) are also material, and although both lie outside the scope of this thesis, they motivate its limitations and future work. Together these strands sit within the broader agenda of aligning artificial intelligence with climate-change mitigation (Kaack et al., 2022).

2.2 Carbon-Intensity Signals

Exploiting the when-and-where sensitivity identified above first requires a signal that quantifies how clean electricity is at a given time and place. The most common such signal is grid carbon intensity (CI), the average emissions per unit of electricity consumed, which is published at hourly granularity and used directly as a scheduling input in production systems (Radovanovic et al., 2023). Because CI is an average measure, a more marginal alternative has been proposed in the form of locational marginal emissions, which attribute emissions to the

specific generator that responds to an incremental load and can differ materially from average CI in large grids (Cote & Sun, 2025; Lindberg et al., 2022). A different signal again arises from the 24/7 carbon-free energy (CFE) accounting movement, which matches each hour of consumption against carbon-free generation in the same hour rather than against annual renewable certificates (Riepin & Brown, 2024). This thesis draws on both CI and the carbon-free energy fraction, using the latter to distinguish electricity that is clean because of genuine renewable supply from electricity that is only incidentally low-carbon.

Because schedules must be planned before the relevant hour arrives, a supporting literature forecasts carbon intensity using neural, ensemble, and uncertainty-aware methods (A. Li, Liu, & Ding, 2024; Maji, Shenoy, & Sitaraman, 2022; Yan, Wang, Liu, & Ding, 2025). The present work treats CI and CFE as given inputs and isolates the scheduling problem, so forecasting is complementary to, rather than part of, the optimization model developed here.

2.3 Temporal Workload Shifting

With a carbon signal in hand, the first way to act on it is temporal, deferring delay-tolerant work to hours of lower carbon intensity. This lever was characterized in detail by Wiesner, Behnke, Scheinert, Gontarska, and Thamsen (2021), who quantified achievable savings for delay-tolerant workloads across several national grids and found them to depend strongly on the region and on the available delay budget. The most prominent production realization is Google’s carbon-aware computing platform, which shifts flexible batch load across the day using Virtual Capacity Curves as a scheduling interface and reports meaningful operational-carbon reductions at scale without violating service objectives (Radovanovic et al., 2023). Building on this idea, subsequent systems work has operationalized temporal shifting through cluster schedulers for machine-learning workloads (Xu, Sun, Tian, Zhang, & Chen, 2025), elastic resource scaling that varies capacity across carbon windows (Hanafy, Wu, Irwin, & Shenoy, 2025), and system-level facilities that cap or shape application-level emissions (Souza, Bashir, et al., 2023; Thiede, Bashir, Irwin, & Shenoy, 2023).

Despite these advances, a recurring finding qualifies temporal shifting and motivates the present thesis. Several studies show that shifting in time alone yields limited and context-dependent savings, because the benefit shrinks once the delay budget exceeds a few hours and is small in grids whose carbon intensity varies little within a day (Sukprasert et al., 2024; Wiesner et al., 2021).

2.4 Spatial Workload Routing

Because temporal flexibility alone is bounded, attention turns to the spatial lever, which routes flexible work to cleaner regions. Its foundations lie in the geographical load-balancing literature, which established that distributing load across locations according to local grid conditions can reduce both cost and emissions (Lindberg et al., 2022; Liu et al., 2012). More recent carbon-aware variants route requests to data centers with cleaner electricity subject to latency limits (Souza, Jasoria, et al., 2023), extend the idea to federated and geo-distributed machine learning (Bian, Wang, Ren, & Xu, 2024; Moore et al., 2025), and add fairness so that no single region bears a disproportionate environmental burden (P. Li, Yang, Wierman, & Ren, 2024).

A closely related strand goes further and optimizes the spatial and temporal dimensions jointly, which is the strand this thesis contributes to. The most directly comparable work is that of Riepin, Brown, and Zavala (2025), who formulate a linear program for spatio-temporal load shifting across European data centers; their objective, however, is 24/7 CFE matching through dedicated procurement rather than dispatch against real-time grid carbon intensity, and

they report an optimal data-center separation of three to four hundred kilometres. Other joint formulations adopt mixed-integer models to add multi-resource footprints (Attenni, Moawad, Bartolini, & Thamsen, 2025) or precedence-constrained workloads (Lechowicz, Shenoy, et al., 2025). Even so, a recent review concludes that genuine joint spatio-temporal optimization remains comparatively rare and that most methods still address one dimension at a time (Asadov, Coroamă, Franzil, Galantino, & Finkbeiner, 2025). The present thesis differs from the joint work that does exist in two respects, in that it uses real multi-year grid CI and CFE signals as the dispatch basis and decomposes the realized saving into its spatial and temporal parts.

2.5 Heuristic versus Optimization-Based Scheduling

Whether they exploit temporal, spatial, or joint flexibility, the approaches surveyed so far divide along a methodological line that constitutes the second gap this thesis addresses. On one side, heuristic methods such as greedy placement, threshold-based deferral, and rule-based routing dominate deployed systems because they are fast and simple to operate (Acun et al., 2023; Radovanovic et al., 2023; Wijayawardana & Chien, 2025). A growing body of work strengthens these heuristics with formal guarantees, as online and learning-augmented algorithms provide competitive bounds under uncertain inputs (Bostandoost et al., 2024; Lechowicz, Christianson, et al., 2025), meta-algorithms select among heuristics adaptively (Bostandoost et al., 2025), and distributionally robust or game-theoretic formulations offer probabilistic performance guarantees across a network of data centers (Breukelman, Hall, Belgioioso, & Dörfler, 2024; Hall, Micheli, Belgioioso, Radovanović, & Dörfler, 2024).

On the other side, optimization-based approaches solve directly for a provably optimal schedule. Convex and linear programs were used early on to co-optimize workload and cooling (Liu et al., 2012) and, more recently, to match clean energy across space and time (Riepin et al., 2025) and to minimize cost and carbon jointly across data-center micro-grids (Hao, Liu, & Deng, 2024). An exact optimum is valuable in this setting for two distinct reasons. It provides an upper bound against which any heuristic can be measured, and it guarantees that operational constraints are satisfied by construction. Bridging the two traditions, holistic design frameworks such as Carbon Explorer embed greedy scheduling within a larger exploration of operational and embodied trade-offs (Acun et al., 2023), while open testbeds increasingly support empirical comparison of the competing methods (Wiesner, Behnke, Kilian, Steinke, & Kao, 2025).

2.6 Positioning of This Thesis

Taken together, the literature reviewed above leaves a specific gap that this thesis addresses. Most deployed carbon-aware schedulers are heuristic and therefore cannot certify how close they come to optimal (Acun et al., 2023; Radovanovic et al., 2023); most schedulers that exploit both temporal and spatial flexibility do so empirically or treat the two dimensions in isolation (Asadov et al., 2025; Sukprasert et al., 2024); and the closest exact joint formulation optimizes for dedicated 24/7 CFE procurement using synthetic generation profiles rather than dispatch against real grid signals (Riepin et al., 2025). In response to these three observations, this thesis formalizes carbon-aware scheduling as a deterministic linear program that is provably optimal by construction, optimizes the temporal and spatial dimensions jointly, draws on real two-year hourly CI and CFE data across five structurally diverse grid regions, and decomposes the realized saving to attribute it to spatial versus temporal flexibility. The resulting model is intended not as a replacement for the operational heuristics that dominate practice, but as the optimal reference they have so far lacked and as an instrument for understanding which form of flexibility actually drives carbon savings.

3 Data and Exploratory Analysis

This section introduces the carbon data that drive the model, the five grid regions over which it operates, and the exploratory analysis that motivates the modelling choices formalized in Section 4. It first describes the data sources and the rationale for the regional selection, and then characterizes the spatial and temporal structure of carbon intensity to establish where carbon savings can be found.

3.1 Data and Target Regions

The data consist of hourly grid measurements from ElectricityMaps over the full 2024–2025 period, giving 17,544 hours per region across five regions. The regions are chosen so that the conclusions do not depend on any single grid, spanning clean to high-carbon supply and variable to flat carbon intensity. PJM (US-MIDA-PJM) is a large coal-and-gas grid with high carbon intensity and large intra-day swings; NYISO (US-NY-NYIS) is cleaner and more moderate, drawing on hydro and nuclear; Finland (FI) is among the cleanest, combining hydro, nuclear, and wind; Belgium (BE) sits in the middle, with substantial nuclear and a diurnal solar signal; and Singapore (SG) serves as a control region, since its almost entirely gas-fired supply gives it high but flat carbon intensity, which tests whether carbon-aware scheduling helps where the grid does not vary.

Two hourly signals drive the model, both available at the resolution at which the scheduling decision is made. The first is carbon intensity (CI), the average grams of CO₂-equivalent emitted per kilowatt-hour, retrieved from the `carbonIntensity` field. The second is the carbon-free energy fraction (CFE), the share of consumed electricity supplied by carbon-free sources, computed from the `powerBreakdown` field as the combined contribution of wind, solar, hydro, and nuclear generation.

The two signals are retained separately because they measure different things. Carbon intensity measures how much carbon a kilowatt-hour emits on average, whereas CFE measures how structurally clean the supply is. Two hours can share the same CI for different reasons, one because wind is displacing fossil generation and another because demand is low while fossil plants still run, and only CFE distinguishes them. Using both lets the model favour electricity that is clean because of renewable supply, consistent with 24/7 carbon-free accounting (Riepin & Brown, 2024). The regions also differ in carbon-free composition, with nuclear material in Finland and Belgium, where a CFE-aware objective differs most from one based on CI alone. The thesis uses average CI rather than locational marginal emissions, which attribute carbon to the marginal generator and can diverge from average CI in large grids; this choice is revisited as a limitation in Section 6 (Cote & Sun, 2025; Lindberg et al., 2022).

3.2 Exploratory Analysis

Before the model is specified, several features of the data establish where carbon savings can be found and justify the modelling choices that follow. The first concerns the spatial structure of carbon intensity, which matters in two distinct ways. The more visible is the spread in levels across regions. As Figure 1 shows, median CI ranges from roughly 80 gCO₂eq/kWh in Finland to more than 530 in Singapore, a difference of close to a factor of seven. A spread of this magnitude means that routing flexible load from a dirty region to a clean one can cut its carbon by more than any rescheduling within a single region could, a first indication that the spatial dimension is likely to dominate. Table 1 reports the per-region descriptive statistics referred to throughout this section.

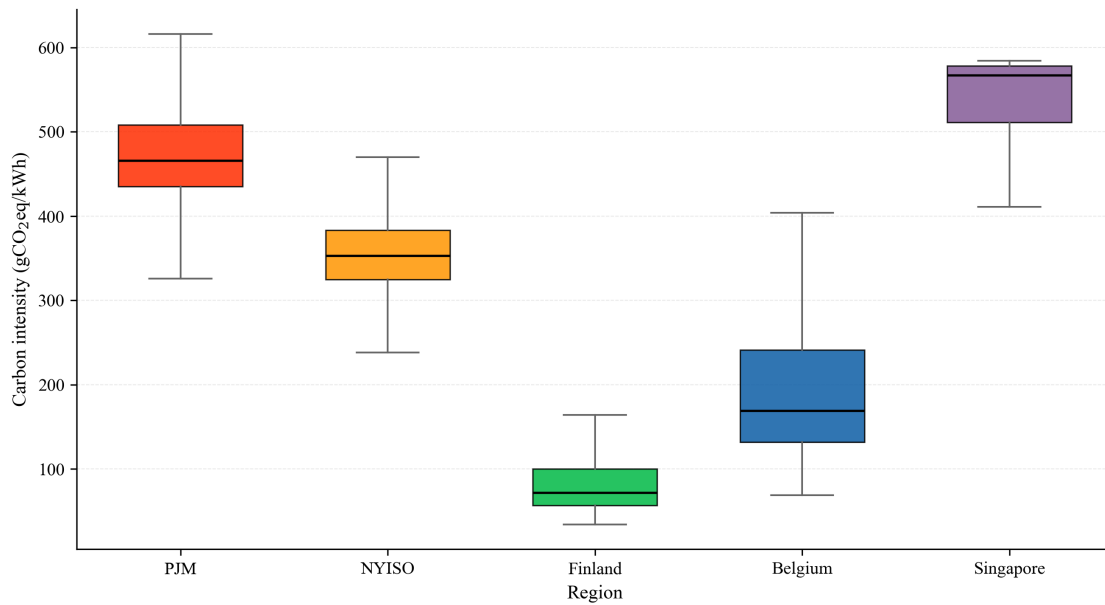


Figure 1: Distribution of hourly carbon intensity by region over 2024–2025. Median CI differs by almost a factor of seven between the cleanest region (Finland) and the highest-carbon one (Singapore), establishing a large headroom for spatial routing. Singapore combines high CI with a narrow spread, foreshadowing its role as a control region.

Region	Grid zone	CI mean	CI sd	CI range	CFE (%)	R^2
PJM	US-MIDA-PJM	472	52	313–627	42.0	0.748
NYISO	US-NY-NYIS	354	44	215–499	51.1	0.768
Finland	FI	83	35	34–285	94.0	0.968
Belgium	BE	191	76	69–433	76.1	0.958
Singapore	SG	533	60	391–584	2.7	0.059

Table 1: Descriptive statistics of the hourly data per region over 2024–2025 (17,544 hours each). CI columns (mean, standard deviation, and min–max range) are in gCO₂eq/kWh; CFE is the mean carbon-free energy fraction; and R^2 is the share of CI variance explained by CFE. The clean European grids (Finland, Belgium) combine low CI with high spread, contrasting with the high-carbon, low-spread control (Singapore).

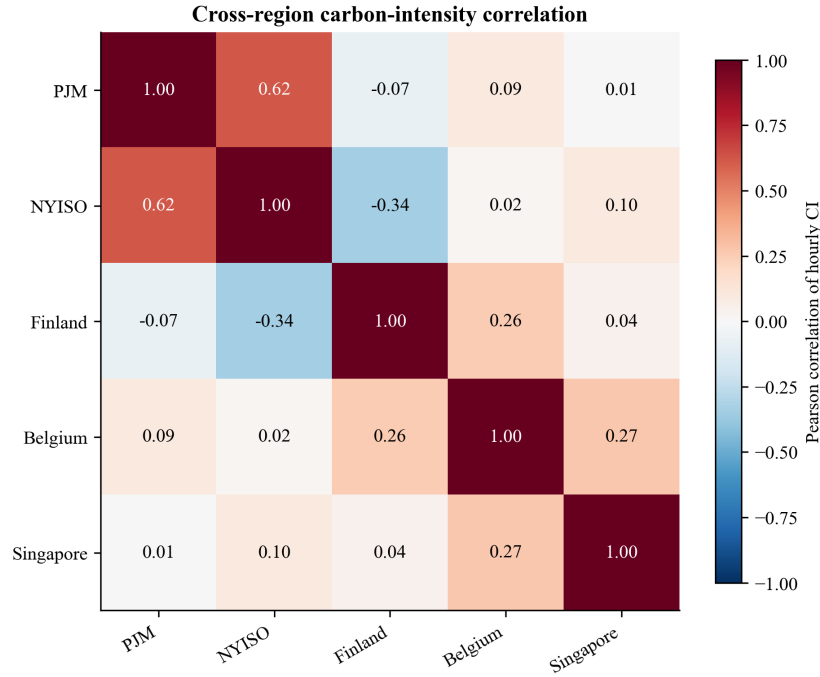


Figure 2: Pearson correlation of hourly carbon intensity between regions over 2024–2025. Only the two US grids are strongly correlated; every other pair is near zero or negative. This decorrelation is what makes spatial routing effective, since a clean region is almost always available when another is dirty.

A difference in levels is necessary but not sufficient for spatial routing to help; what matters equally is whether the regions move together. Figure 2 reports the Pearson correlation of hourly CI between every pair of regions. With the exception of the two US grids, which share a generation mix and correlate at 0.62, the off-diagonal correlations are close to zero or negative, including a correlation of -0.34 between NYISO and Finland. This spatial decorrelation is the structural precondition for geographic load shifting, because regions that rarely peak together ensure that when one grid is carbon-intensive another is usually clean and able to absorb the load. Without such decorrelation, routing would add little to what temporal shifting already provides (Lindberg et al., 2022; Riepin et al., 2025).

The second feature is the temporal structure of carbon intensity within the day, shown in Figure A.1. The size of the intra-day swing varies sharply across regions. Belgium has the largest diurnal range, driven by a midday solar trough, whereas Finland is almost flat on an hour-of-day basis. This low diurnal range is misleading, however, because Finland has the highest overall variability in the dataset once day-to-day fluctuations are included, as its monthly profile in Figure 3 shows. The distinction matters for modelling, since Finland’s variability comes from multi-day wind patterns rather than a daily cycle, so the value of temporal shifting depends on the planning window and the source of variation, not on the time of day alone. This region-dependence of temporal savings is consistent with prior empirical findings (Sukprasert et al., 2024; Wiesner et al., 2021).

Variation within the day is only part of the temporal picture. Figure 3 shows mean CI by calendar month, which exposes a seasonal structure that the diurnal view hides. Finland is cleanest in late spring and summer, when hydro and wind peak and its mean CI falls below $60 \text{ gCO}_2\text{eq/kWh}$, while Belgium softens through the warmer months as solar contributes and PJM and Singapore stay high year-round. In the clean European grids these seasonal swings are larger than the diurnal ones, which is why a full two-year horizon, rather than a single representative week, is needed to capture the temporal opportunity, and why the results in

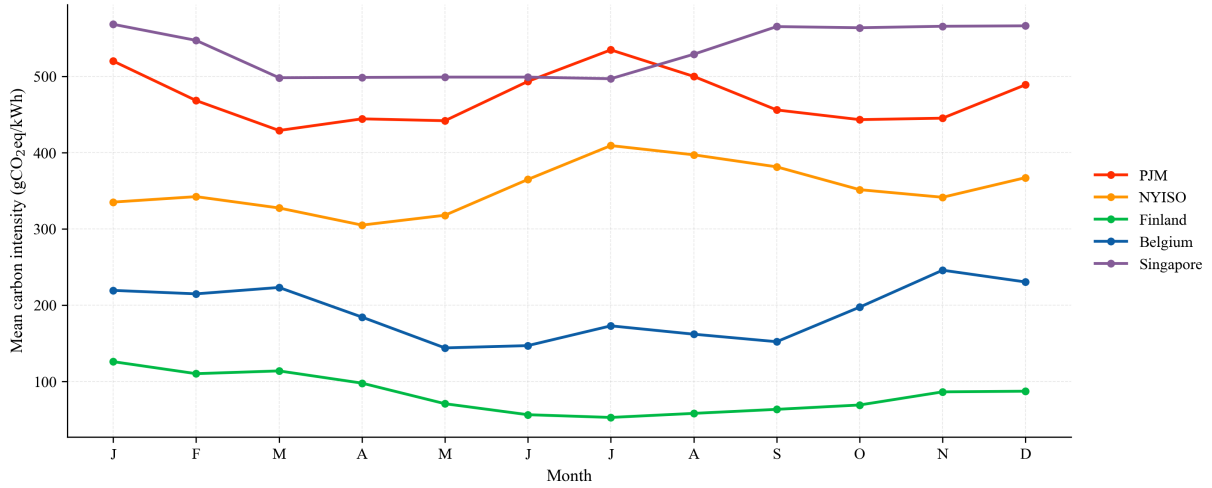


Figure 3: Mean carbon intensity by calendar month per region over 2024–2025. Finland and Belgium show pronounced seasonal troughs in the warmer months as renewable generation peaks, whereas PJM and Singapore remain high throughout the year. These seasonal swings exceed the diurnal ones in the clean grids, motivating the two-year evaluation horizon.

Section 5 are reported season by season.

The third feature concerns which signal should drive the model. Figure 4 regresses carbon intensity on the carbon-free energy fraction in each region. In the four grids with meaningful renewable or nuclear supply, CFE explains most of the variation in CI, with the coefficient of determination reaching 0.97 in Finland and exceeding 0.94 in Belgium. The exception is Singapore, where CFE is near zero throughout and explains almost none of the variation, as expected for an almost entirely gas-fired grid. The strength of this relationship justifies entering CFE into the objective as a structural-cleanliness discount, and the steepness of the fitted slopes implies a large discount weight, which informs the choice of the corresponding parameter in Section 4.

The regression shows that CFE tracks CI, but not why CFE is preferable to a simpler renewables-only signal. Figure A.2 resolves this by decomposing each region’s carbon-free supply into renewable and nuclear shares. The two clean European grids reach high CFE by different routes, Finland through a large renewable share supplemented by nuclear and Belgium through a near-even split, whereas PJM’s moderate 42 % CFE is almost entirely nuclear, with renewables contributing under 8 %. Because nuclear is carbon-free yet invisible to a renewables-only measure, a region like PJM would appear far dirtier than it is under a renewable-fraction objective. This is the empirical reason the model uses the carbon-free energy fraction rather than the renewable fraction alone.

Taken together, the descriptive statistics in Table 1 and the figures preview the central empirical finding. The regions differ markedly in carbon level, which creates the headroom that spatial routing exploits, while only Finland and Belgium combine that level difference with substantial within-region variability, so the temporal lever is expected to matter chiefly in those two grids. Singapore, high in carbon yet flat and with no carbon-free supply, is expected to benefit from neither lever, which is what makes it a useful control.

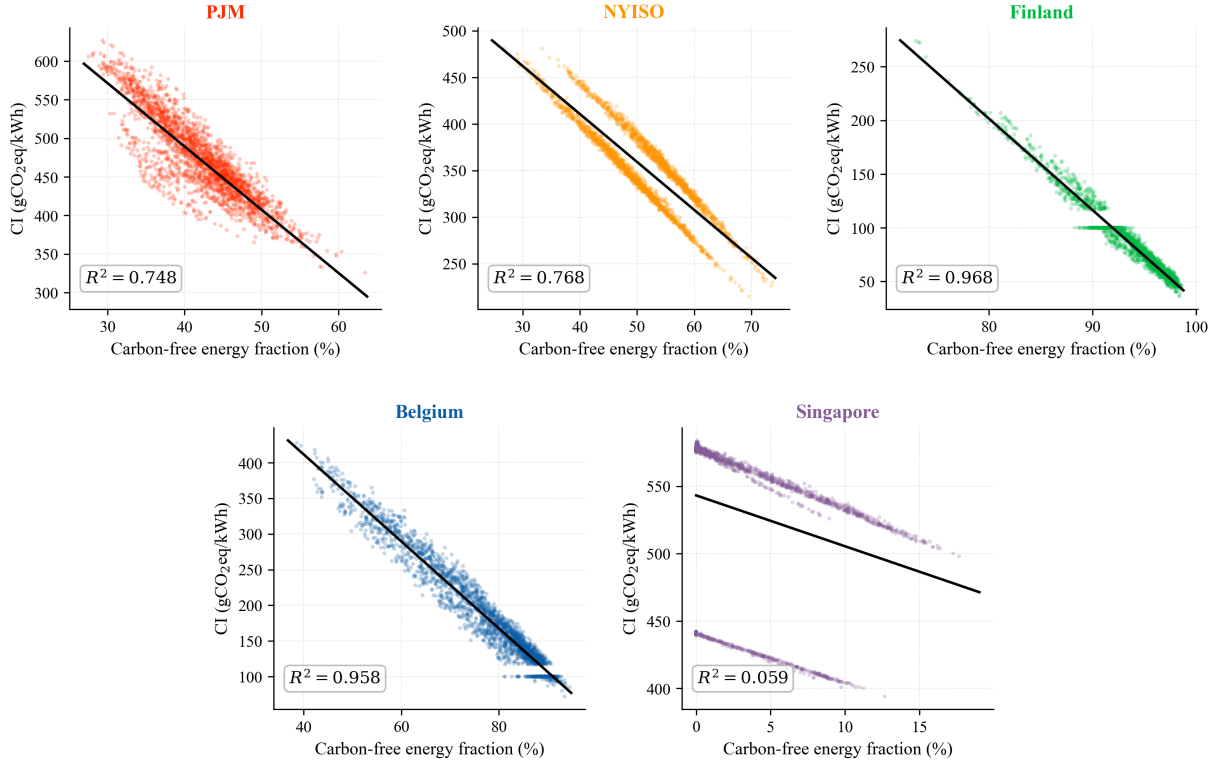


Figure 4: Carbon intensity against the carbon-free energy fraction, with an ordinary-least-squares fit per region. The carbon-free energy fraction explains most of the variation in carbon intensity in every region except gas-dominated Singapore, supporting its use as a cleanliness signal alongside carbon intensity.

4 Optimization Model

This section presents the optimization model. Figure 5 gives an overview of how it operates, after which the section states the scheduling problem, defines the inputs and the decision variable, the objective, and the seven constraints, and closes with the parameter settings and the solution method.

The model maps the hourly grid signals and flexible demand to a schedule through a linear program that minimizes the operational carbon of the schedule, discounted by the carbon-free energy fraction, together with an equity term, subject to the seven operational constraints. Its solution is an optimal schedule specifying how much flexible load runs in each region and hour. Applied in a rolling window across the two-year horizon, the schedule is then evaluated against a no-optimization baseline and three reference schedulers, and analysed through the sensitivity and decomposition studies of Section 5. The remainder of this section defines each component in turn.

4.1 Problem Setting

A data-center operator receives a stream of flexible compute demand that must be completed within a bounded deadline rather than at the instant it arrives. The operator runs facilities in several regions, each connected to a distinct electricity grid, and must decide how much of the demand to serve in each region during each hour of a planning horizon. Serving a unit of load in a given region-hour emits carbon in proportion to that grid’s carbon intensity at that hour, so the same unit of work carries a different carbon cost depending on when and where it runs.

Two degrees of freedom make this cost partly controllable. The operator can defer flexible

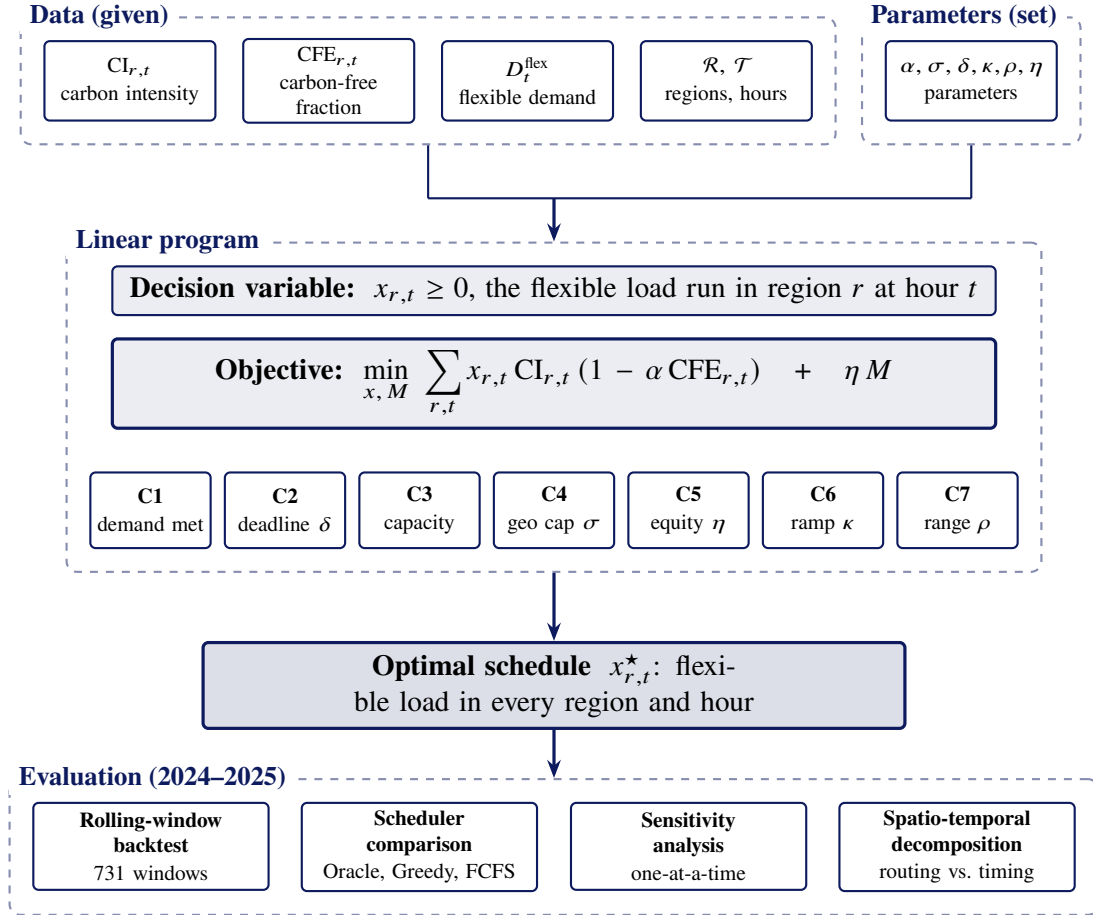


Figure 5: Overview of the optimization model. Hourly carbon signals, flexible demand, and operational parameters enter a linear program defined by a single decision variable $x_{r,t}$, a carbon-minimizing objective discounted by the carbon-free energy fraction, and seven operational constraints (C1–C7). Its solution is the optimal schedule $x_{r,t}^*$, which is evaluated over the 2024–2025 horizon through a rolling-window backtest, a comparison against three reference schedulers, a one-at-a-time sensitivity analysis, and a spatio-temporal decomposition.

load to later hours within the deadline, which constitutes temporal flexibility, and it can move load from its home region to a cleaner one, which constitutes spatial flexibility. Neither degree of freedom is unconstrained in practice, because a deferral deadline limits how long work can wait, a geographic cap limits how much load may leave the home region, and ramp and capacity limits constrain how sharply a region’s load can change between consecutive hours. The problem is therefore to choose a complete assignment of flexible load to region-hours that minimizes total emitted carbon while respecting these operational limits, solved to optimality rather than through a heuristic rule.

4.2 Decision Variables and Objective Function

Let $\mathcal{R} = \{1, \dots, R\}$ be the set of grid regions, one of which, r_0 , is the home region where demand originates, and let $\mathcal{T} = \{1, \dots, T\}$ be the set of hours in the planning horizon, indexed by t . For each region-hour the carbon intensity $CI_{r,t}$ and the carbon-free energy fraction $CFE_{r,t} \in [0, 1]$ are known inputs drawn from the data of Section 3. Flexible demand arrives over time, with D_τ^{flex} kilowatt-hours arriving at hour τ and a total of $D^{\text{flex}} = \sum_{\tau \in \mathcal{T}} D_\tau^{\text{flex}}$.

The single decision variable is the flexible load assignment

$$x_{r,t} \geq 0, \quad r \in \mathcal{R}, t \in \mathcal{T}, \quad (1)$$

which gives the kilowatt-hours of flexible work served in region r during hour t . The matrix $\mathbf{x} \in \mathbb{R}^{R \times T}$ therefore encodes a complete schedule, specifying jointly *when* (the column t) and *where* (the row r) each unit of load runs. A few auxiliary variables linearize two of the constraints below, namely $M \geq 0$ for the worst-case regional carbon (C5) and per-region bounds u_r and ℓ_r for the load range (C7); each is defined where it first appears.

Given these variables, the model minimizes a weighted sum of three terms,

$$\min_{\mathbf{x}, M} \underbrace{\sum_{r \in \mathcal{R}} \sum_{t \in \mathcal{T}} x_{r,t} \text{CI}_{r,t} (1 - \alpha \text{CFE}_{r,t})}_{\text{operational carbon}} + \underbrace{\gamma \sum_{r \neq r_0} \sum_{t \in \mathcal{T}} x_{r,t}}_{\text{transfer cost}} + \underbrace{\eta M}_{\text{equity}}. \quad (2)$$

The first term is the operational carbon of the schedule. Rather than weighting each unit of load by raw carbon intensity, it applies a multiplicative discount $(1 - \alpha \text{CFE}_{r,t})$ so that electricity which is structurally clean, in the sense of a high carbon-free energy fraction, is treated as cheaper than its carbon intensity alone would suggest. The coefficient $\alpha \in [0, 1]$ controls the strength of this discount. At $\alpha = 0$ the discount vanishes and the model minimizes pure carbon intensity, whereas at $\alpha = 1$ a fully carbon-free hour carries zero effective cost. The exploratory regression in Section 3 found implied discounts of 0.83 to 1.00 across the four non-fossil grids, so the signal is empirically strong, and the model adopts a conservative $\alpha = 0.5$ as its baseline.

The second term charges a per-unit cost γ for any load routed away from the home region, representing the operational friction of moving work across sites. Because the experiments instead control geographic movement directly through the hard cap in constraint C4, γ is set to zero throughout, and the term is retained only to show that a soft transfer penalty is expressible within the same formulation. The third term penalizes the carbon borne by the most heavily loaded region through the weight η , which discourages concentrating emissions in a single grid. Setting γ and η to zero recovers a pure carbon-minimization objective, which makes the contribution of each term independently testable.

4.3 Constraints

The schedule must satisfy seven operational constraints, presented in turn below and referred to throughout as C1 to C7. The last two are written in the linear two-inequality form in which they are actually solved.

The first constraint is demand completion. All flexible demand must eventually be served, so that the model lowers carbon by reshaping the schedule rather than by dropping work,

$$\sum_{r \in \mathcal{R}} \sum_{t \in \mathcal{T}} x_{r,t} = D^{\text{flex}}. \quad (\text{C1})$$

The second is a latency bound. A batch of demand arriving at hour τ must be completed within δ hours of arrival, which stops the model from deferring everything to a single distant clean hour at the cost of every deadline,

$$\sum_{r \in \mathcal{R}} \sum_{t'=\tau}^{\tau+\delta} x_{r,t'} \geq D_{\tau}^{\text{flex}} \quad \forall \tau \in \mathcal{T}. \quad (\text{C2})$$

The third is a pair of capacity bounds. The load served in each region in any hour lies between a floor and a ceiling fixed by its physical supply and cooling limits,

$$C_r^{\min} \leq x_{r,t} \leq C_r^{\max} \quad \forall r \in \mathcal{R}, t \in \mathcal{T}. \quad (\text{C3})$$

Because the floor satisfies $C_r^{\min} \geq 0$, non-negativity of $x_{r,t}$ follows automatically.

The fourth constraint, which proves the most consequential in the results, is the geographic transfer limit. At each hour the load served outside the home region cannot exceed a fraction σ of the total dispatched load,

$$\sum_{r \neq r_0} x_{r,t} \leq \sigma \sum_{r \in \mathcal{R}} x_{r,t} \quad \forall t \in \mathcal{T}. \quad (\text{C4})$$

Setting $\sigma = 1$ removes the restriction, whereas smaller values progressively confine work to the home grid. Expressing the bound relative to dispatched load, rather than to total demand, keeps it meaningful however much demand is deferred.

The fifth constraint is the equity auxiliary. The variable M is held above every region's total carbon, so that the term ηM in the objective acts on the most-burdened region,

$$M \geq \sum_{t \in \mathcal{T}} x_{r,t} \text{CI}_{r,t} \quad \forall r \in \mathcal{R}. \quad (\text{C5})$$

The sixth constraint is a ramp-rate limit, which smooths the schedule in time by capping how sharply a region's load may change between consecutive hours. The absolute value is expressed as the two linear inequalities actually solved, bounding upward and downward steps by the same fraction κ of capacity,

$$x_{r,t} - x_{r,t-1} \leq \kappa C_r^{\max}, \quad (\text{C6a})$$

$$x_{r,t-1} - x_{r,t} \leq \kappa C_r^{\max}, \quad (\text{C6b})$$

for all $r \in \mathcal{R}$ and $t \in \{2, \dots, T\}$.

The seventh constraint is a dynamic-range limit, which caps the peak-to-trough spread of a region's load over the whole horizon. It is linearized with per-region auxiliary variables u_r and ℓ_r that bound the highest and lowest hourly load, so that the range itself is capped at a fraction ρ of capacity,

$$\ell_r \leq x_{r,t} \leq u_r \quad \forall r \in \mathcal{R}, t \in \mathcal{T}, \quad (\text{C7a})$$

$$u_r - \ell_r \leq \rho C_r^{\max} \quad \forall r \in \mathcal{R}. \quad (\text{C7b})$$

The ramp and range limits are inactive at $\kappa = 1$ and $\rho = 1$ and tighten as those values fall, following the capacity-variation characterization of [Wijayawardana and Chien \(2025\)](#).

4.4 Parameters and Solution Method

The objective and constraints above introduce seven tunable coefficients, α , σ , δ , κ , ρ , η , and γ , which are distinct from the data inputs of Section 4.2 in that they are chosen by the modeller rather than read from the grid. Table 2 summarizes each coefficient, its role, and the range over which it is tested in Section 5. Two parameter regimes recur in the evaluation. The sensitivity analysis varies one parameter at a time while holding the others at their loosest, carbon-permitting settings, which isolates the marginal effect of each. The scheduler comparison

Symbol	Role	Interpretation	Tested range
α	CFE discount	Weight on the carbon-free energy signal	{0, 0.25, 0.5, 0.75, 1.0}
σ	Geographic cap	Max share of load served off the home region	{0.3, 0.5, 0.7, 0.9, 1.0}
δ	Latency window	Hours within which arriving load must finish	{6, 12, 24, 48}
κ	Ramp rate	Max hour-to-hour change, as a share of $C^{\max}$	{0.1, 0.2, 0.5, 0.8, 1.0}
ρ	Dynamic range	Max peak-to-trough range, as a share of $C^{\max}$	{0.2, 0.4, 0.6, 0.8, 1.0}
η	Equity weight	Penalty on the most-burdened region's carbon	{0, 0.1, 0.3, 0.5, 1.0}
γ	Transfer cost	Per-unit cost of off-home routing (set to 0)	—

Table 2: Model parameters, their role, and the ranges tested in the sensitivity analysis. The geographic transfer cost γ is held at zero because geographic movement is controlled directly by the hard cap C4.

instead fixes a single fully constrained operational regime, $\sigma = 0.6$, $\kappa = 0.5$, and $\rho = 0.7$, that represents a realistic deployment in which all limits are active at once.

With the full model in place, solving it is straightforward. The program is linear in the continuous variables $\mathbf{x}$, M , and the range bounds u_r and ℓ_r . The only operations that are nonlinear in spirit, the absolute value in the ramp limit and the spread in the range limit, were already written above as the linear inequality pairs C6a–C6b and C7a–C7b, so the program is linear as stated and needs no further reformulation. It is solved with the HiGHS solver (Huangfu & Hall, 2018) through `scipy.optimize.linprog` (Virtanen et al., 2020), with Gurobi available as an interchangeable backend. Because the formulation is a linear relaxation over continuous load, it solves in well under a second per instance even at the full horizon of $R = 5$ regions and $T = 17,544$ hours, so no integer variables are required unless workloads must be dispatched in discrete jobs.

The two-year evaluation in Section 5 applies the model in a rolling-window backtest, solving the program independently on each successive scheduling window across 2024 and 2025 and accumulating the realized carbon. To benchmark the optimal schedule against a fast rule of the kind used in practice, a greedy heuristic is also implemented. At each window it sorts region-hour pairs by their effective cost $CI_{r,t}(1 - \alpha CFE_{r,t})$ and fills the cheapest feasible pairs first, subject to the capacity and geographic limits but without solving an optimization problem. The greedy rule therefore respects the geographic cap but not the ramp or range constraints, a distinction that becomes important when its carbon is compared against the LP in Section 5.

5 Computational Results

This section reports the empirical evaluation of the model. It first validates the formulation, through a behavioural perturbation test and an integration check on real data, and then presents the results of the two-year backtest, covering a representative schedule, the sensitivity analysis, seasonal behaviour, the comparison of solution methods, and a decomposition of where the carbon savings come from. Unless stated otherwise, results are computed over the full 2024–

Perturbation	Observed response	Result
CI shock: Finland's CI raised from 52–55 to 700, above PJM	Finland 6.0 $\rightarrow$ 0.0 kWh; Belgium 0.0 $\rightarrow$ 6.0	PASS
Demand raised from 10 to 18 kWh	All served (C1 exact); Finland rises to 10.8	PASS
Deferral window δ shortened from 6 h to 1 h	Carbon 2,574 $\rightarrow$ 2,808 gCO ₂ eq	PASS
Finland capacity squeezed mildly ($C_{\max}$ 10 $\rightarrow$ 3 kWh/h)	Finland total unchanged at 6.0; spread out	PASS
Finland capacity squeezed severely ($C_{\max}$ 10 $\rightarrow$ 0.5)	Finland total 6.0 $\rightarrow$ 3.0; shortfall rerouted	PASS

Table 3: Perturbation test on the three-region, six-hour instance with all seven constraints active. Thirteen individual checks across these five scenarios all pass.

2025 horizon, which the rolling-window backtest divides into 731 scheduling windows.

5.1 Model Validation

Validation proceeds in two parts that answer different questions. The first asks whether the model reacts correctly when an input changes, and the second asks whether the full formulation holds together on real data.

The perturbation test addresses the first question on a small constructed instance of three real regions (PJM as home, Finland, and Belgium) over a six-hour window, with all seven constraints active so that the correct response to each change is known in advance. One input is changed at a time and the schedule is checked against its predicted reaction. Table 3 reports the outcome of each scenario and Figure A.3 shows the corresponding hourly schedules. Every metric moves in the predicted direction, in that load reroutes as soon as a region ceases to be the cleanest option, additional demand flows to the preferred region first, a shorter deferral window forces load out of the clean hours and raises carbon, and a tighter hourly cap reshapes the schedule before it reduces throughput. All thirteen individual checks across the five scenarios pass.

The integration check addresses the second question by running the complete model once on the first 168 hours of real data, again with all seven constraints active. Table 4 reports, for each constraint, the observed value against its limit and whether it binds. Every constraint is satisfied within numerical tolerance, and the pattern of binding is itself informative. Demand completion, the geographic cap, the fairness auxiliary, and the ramp limit actively shape the solution, whereas the latency and dynamic-range limits remain slack because the optimal schedule already respects them with room to spare. Relative to serving all load at the home region, the model reduces carbon by 39.8 % on this instance. Together the two checks establish that the formulation is both logically correct and empirically sound before any aggregate result is read from it.

	Constraint	Observed vs. limit	Binding	Status
C1	Demand	7.000 = 7.0 kWh required	Yes	PASS
C2	Latency	Latest placement at hour 19 of 23	No	PASS
C3	Capacity	$\min x = 0$; $\max x = 0.30$ vs. $C_{\max} = 1.0$	Partial	PASS
C4	Transfer cap	Max off-home fraction = $0.50 = \sigma$	Yes	PASS
C5	Fairness	$M = 1,683 = \text{PJM regional carbon}$	Yes	PASS
C6	Ramp	Max ramp = $0.20 = \kappa C_{\max}$	Yes	PASS
C7	Dynamic range	Max swing = $0.30 < \rho C_{\max} = 0.40$	No	PASS

Table 4: Integration check on 168 hours of real data ($\sigma = 0.5$, $\kappa = 0.2$, $\rho = 0.4$, $\eta = 0.3$, $\delta = 24$ h). All seven constraints hold within tolerance; four bind and two are slack. LP carbon saving versus a home-only baseline is 39.8 %.

5.2 An Illustrative Schedule

Before turning to aggregate savings, Figure A.4 makes the model’s behaviour concrete for a single representative window. The comparison is deliberately between no shifting and LP shifting, rather than between two optimized schedules, because only the no-shift case represents what the system does without the model. The no-shift baseline serves the entire batch immediately at the home region, while the LP routes it to Finland and spreads it over several hours to respect the ramp limit. In this window the no-shift schedule emits 533 gCO₂ and the LP schedule emits 53, a reduction of 90.0 %, and almost all of that gap comes from routing to a cleaner region rather than from retiming within the home grid. This single window is illustrative only; the quantitative claims rest on the full backtest that follows.

5.3 Sensitivity Analysis

To identify which constraints most limit achievable savings, each of the six tunable parameters is varied across its tested range while the others are held at their loosest settings, following the one-at-a-time convention for models with few parameters (Saltelli et al., 2008). Figure 6 presents the result as a tornado diagram (Eschenbach, 1992), with each bar showing the percentage change in LP carbon as the parameter is tightened and the bars sorted by impact.

The dominant finding is that the geographic cap σ governs achievable savings far more than any other parameter. Tightening σ from 1.0 to 0.3 raises LP carbon by 373.7 % relative to the unconstrained baseline, an order of magnitude more than the next largest effect. By comparison the fairness weight η accounts for 20.2 %, the deferral window δ for 8.2 %, and the dynamic-range, ramp, and CFE-discount parameters ρ , κ , and α for 2.9 %, 2.1 %, and 0.2 % respectively. The practical implication is that how much load may move across regions, rather than how it is smoothed in time or how fairness is weighted, is what determines the carbon a deployment can save.

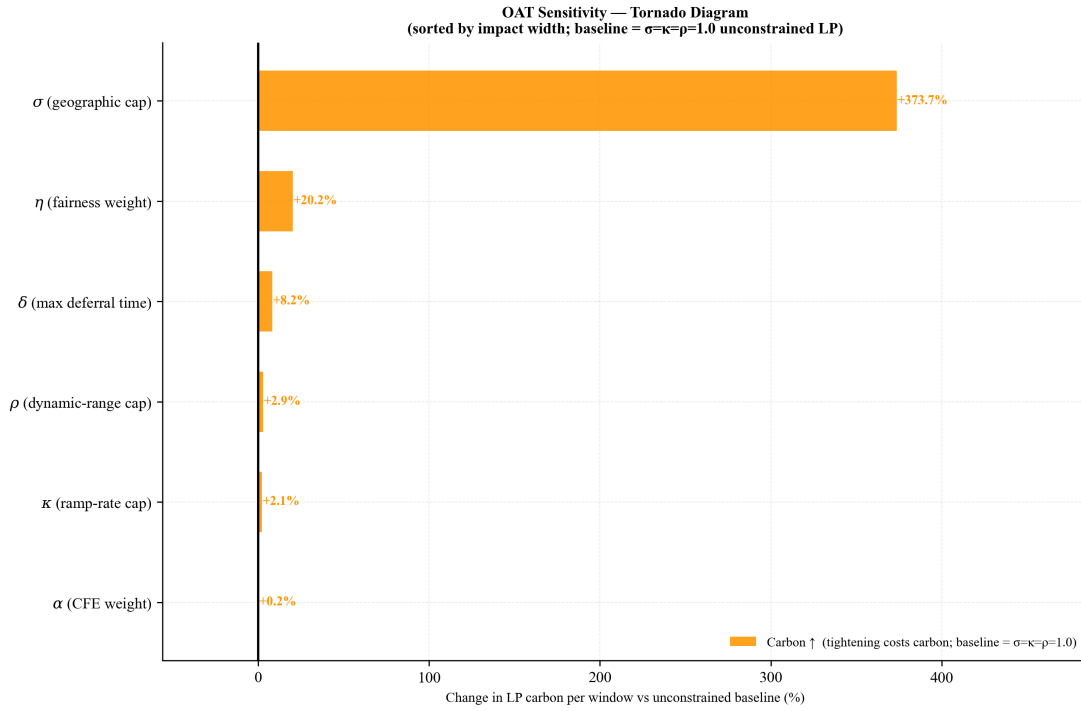


Figure 6: One-at-a-time sensitivity of LP carbon to each parameter, expressed as the percentage change from the unconstrained baseline as the parameter is tightened. The geographic cap σ dominates every other parameter by an order of magnitude.

5.4 Seasonal Analysis

The two-year horizon allows the backtest to be split by season, which tests whether the headline saving depends on the time of year. Figure 7 reports the LP saving and the regional share of allocated load for each season. The saving is high and stable throughout, ranging from 73.4 % in spring to 84.1 % in summer, with winter and autumn at 74.8 % and 81.2 %. Finland absorbs the large majority of scheduled load in every season, and its share rises further in summer as its carbon advantage over the fossil grids widens. The model’s performance therefore rests on one region’s structural carbon advantage, and that advantage holds year-round rather than appearing only in a favourable season.

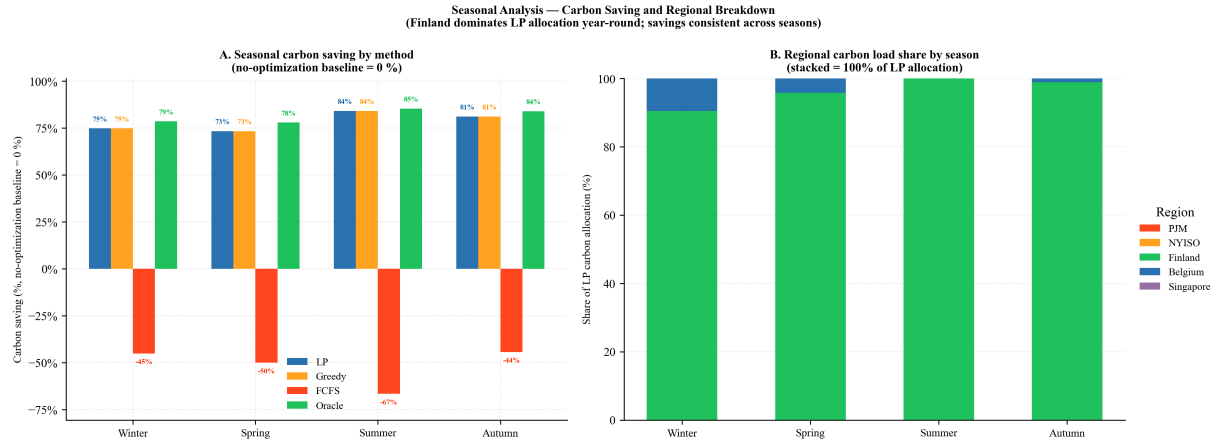


Figure 7: Seasonal carbon saving by method (left) and the regional share of LP-allocated load by season (right). Savings are consistent across seasons, and Finland dominates the allocation throughout, more so in summer as its carbon advantage widens.

5.5 Comparison with Practical Schedulers

Deployed carbon-aware schedulers are heuristic: they apply fast placement rules without solving for a constrained optimum (Acun et al., 2023; Radovanovic et al., 2023). The relevant question is therefore how the exact LP compares with such rules in both carbon and feasibility. Four schedulers are evaluated under the full operational constraints ($\sigma = 0.6$, $\kappa = 0.5$, $\rho = 0.7$), reporting carbon savings against a carbon-agnostic baseline (Acun et al., 2023). Two are exact and serve as references: a perfect-foresight *oracle* that solves the LP over a week-long look-ahead, and the rolling-horizon LP of this thesis. Two are heuristics: a carbon-aware *greedy* rule that fills the cheapest region-hour slots first, and a carbon-agnostic *first-come-first-served* (FCFS) policy. Greedy is used as the heuristic benchmark because, among constructive rules, filling by effective carbon cost is the most direct heuristic for this objective, so its performance bounds that of weaker threshold or routing rules. Table 5 and Figure 8 summarize the comparison.

The oracle saves 36.8 % and serves as a ceiling no deployable method can reach, since it assumes future carbon intensity that is unknown at scheduling time. The LP saves 31.4 %, within a few points of that ceiling, and is the only scheduler that attains this saving while provably respecting every constraint. Greedy saves 31.5 %, within 0.1 percentage points of the LP, but it does so by relaxing the problem: it enforces the geographic cap yet ignores the ramp and range limits, so it optimizes over a larger feasible set than the LP. The near-parity is therefore expected, and it is the carbon price of feasibility rather than evidence that the heuristic is as good. Checking greedy’s hourly schedule against the constraints confirms this: it violates the ramp limit in all 731 backtest windows, with an hour-to-hour swing of 0.60 against a cap of 0.50 (20 % over), while it happens to satisfy the range limit. FCFS saves −51.5 %; it emits about half as much carbon again as the baseline, because it concentrates load in the home region’s consistently high-carbon hours. The ranking is stable across all four seasons.

Scheduler	Saving (%)	Constraints satisfied	Foresight
Oracle (LP, perfect foresight)	36.8	Yes	week ahead
LP (this work)	31.4	Yes	24 h rolling
Greedy (carbon-aware)	31.5	No (ramp)	24 h rolling
FCFS (carbon-agnostic)	-51.5	No	none

Table 5: Carbon saving relative to the no-optimization baseline under the full operational constraints ($\sigma = 0.6$, $\kappa = 0.5$, $\rho = 0.7$). The four schedulers are: *Oracle*, the LP solved with a week of perfect foresight, an idealized upper bound; *LP*, the rolling-horizon model of this thesis; *Greedy*, a carbon-aware heuristic that fills the cheapest region-hour slots first; and *FCFS*, the carbon-agnostic first-come-first-served default. The LP attains near-ceiling savings while satisfying every constraint; greedy matches its carbon only by violating the ramp limit (in all 731 windows); FCFS is worse than no optimization.

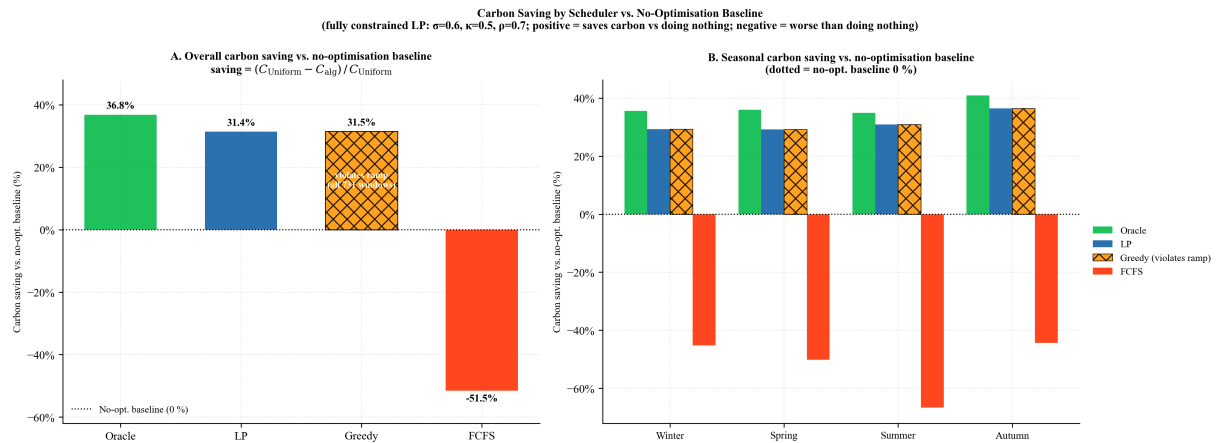


Figure 8: Carbon saving relative to the no-optimization baseline for four schedulers under full operational constraints, overall (left) and by season (right). The oracle is a perfect-foresight ceiling; the LP attains near-ceiling savings while provably respecting every constraint; greedy matches its carbon only by violating the ramp limit; first-come-first-served is worse than doing nothing.

5.6 Spatio-Temporal Decomposition

A final question is whether the headline saving comes from routing load to cleaner regions or from retiming it to cleaner hours. To separate the two, the backtest is re-run under restricted variants of the model, one that may route but not retime, one that may retime but not route, and the full model that may do both. Figure 9 reports the decomposition. Routing alone accounts for almost all of the saving, at 75.6 % against the full model’s 78.4 %, whereas retiming alone is negative, at –35.6 %, because locking all load to the high-carbon home region and optimizing only the hour performs worse than the uniform baseline. The interaction term of 38.4 % shows that routing and retiming are complementary, since choosing the hour is far more valuable once load already sits in a region whose carbon intensity actually varies. In short, where load runs matters far more than when, and timing helps only in concert with routing.

This pattern is explained by the regional structure of the data. Figure A.5 relates each region’s within-region timing saving to the variability of its carbon intensity. Timing yields a meaningful saving only where carbon intensity actually moves, reaching 27.7 % in Belgium and 13.3 % in Finland, the two high-variability grids, while the low-variability grids, PJM, NYISO, and Singapore, gain only six to eight percent from timing regardless of how cleanly it is scheduled. This is consistent with the broader finding that the value of temporal shifting is strongly region-dependent (Sukprasert et al., 2024; Wiesner et al., 2021), and it explains why, with PJM as the home region, retiming alone is unrewarding while routing to a variable clean grid such as Finland or Belgium is not.

6 Conclusion

This thesis formalized carbon-aware data-center scheduling as a deterministic linear program that minimizes operational carbon by jointly choosing when and where flexible load runs across five structurally diverse grid regions. Driven by hourly carbon-intensity and carbon-free-energy signals over a two-year horizon, and subject to seven operational constraints, the model was validated through a behavioural perturbation test and an integration check on real data, and then evaluated in a rolling-window backtest. Under unconstrained flexibility it reduced carbon by up to 78 % relative to a no-optimization baseline, and under realistic operational constraints it saved 31 % while remaining the only method in the comparison that both attained near-ceiling savings and provably respected every constraint.

Beyond the headline saving, two findings carry the main message of the work. First, the sensitivity analysis identified geographic flexibility as the dominant determinant of achievable savings, exceeding every other parameter by an order of magnitude, which indicates that the binding constraint on carbon-aware scheduling is how much load may move across regions rather than how it is shaped in time. Second, the spatio-temporal decomposition showed that routing load to cleaner regions accounts for almost all of the saving, whereas shifting load in time alone, with load confined to a high-carbon home region, can increase carbon. Together these results reframe the practical question for operators from when to run flexible work to where to run it, and they show that the value of an exact optimum lies less in beating a well-tuned heuristic on a given dataset than in providing a provably optimal reference and a guarantee that operational limits are respected.

Several limitations qualify these conclusions and motivate further work. The model uses average carbon intensity rather than locational marginal emissions, which attribute carbon to the generator that responds to an incremental load and can diverge from average intensity in large grids (Cote & Sun, 2025; Lindberg et al., 2022); adopting a marginal signal is a natural

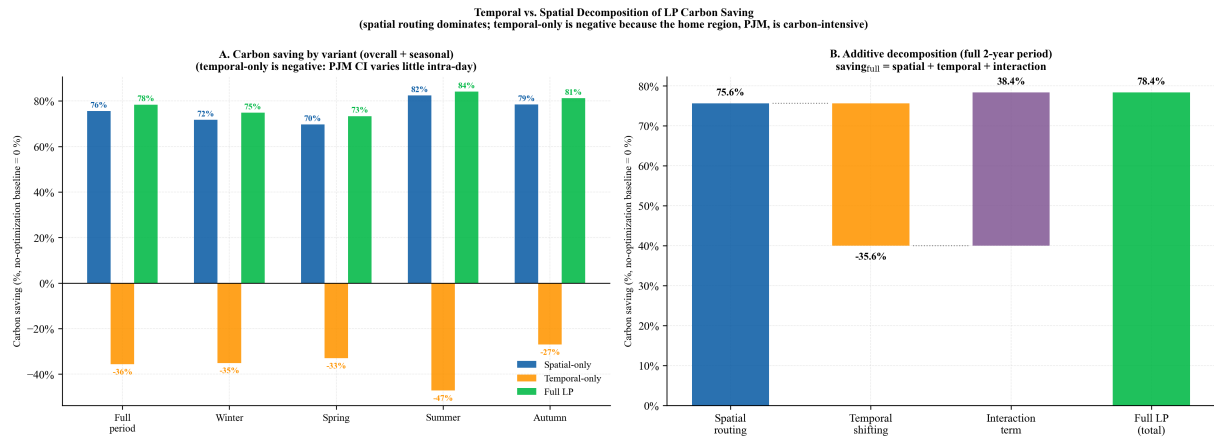


Figure 9: Additive decomposition of the LP saving into routing-only, retiming-only, and interaction components, by season. Routing accounts for almost all of the saving, retiming alone is carbon-increasing, and the two are strongly complementary.

extension. The negative value of time-shifting alone is specific to PJM as the home region, and the cross-region timing comparison rests on a five-point diagnostic rather than a statistical test, so both should be read as indicative rather than conclusive. The formulation is also deterministic and assumes that carbon signals are known over the planning window, whereas a deployment must act on forecasts; a stochastic or distributionally robust extension would quantify the carbon penalty of forecast error (Hall et al., 2024; A. Li et al., 2024). Finally, the model optimizes aggregate load against operational carbon alone, leaving task-level scheduling, embodied hardware emissions, and water consumption outside its scope (Gupta et al., 2021; P. Li et al., 2023).

These limitations suggest a clear agenda for future work, including a marginal-emissions objective, a robust or stochastic treatment of forecast uncertainty, a multi-objective formulation that weighs carbon against water and other footprints, and an extension to precedence-constrained, task-level workloads (Attenni et al., 2025; Lechowicz, Shenoy, et al., 2025). Even within its present scope, however, the model offers a validated and provably optimal account of carbon-aware spatio-temporal scheduling, and a clear empirical answer to which form of flexibility actually drives carbon savings.

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Appendix A Supplementary Figures

The following figures support the exploratory analysis of Section 3 and the results of Section 5. They are referenced from the main text where relevant and collected here to keep the body within the page limit.

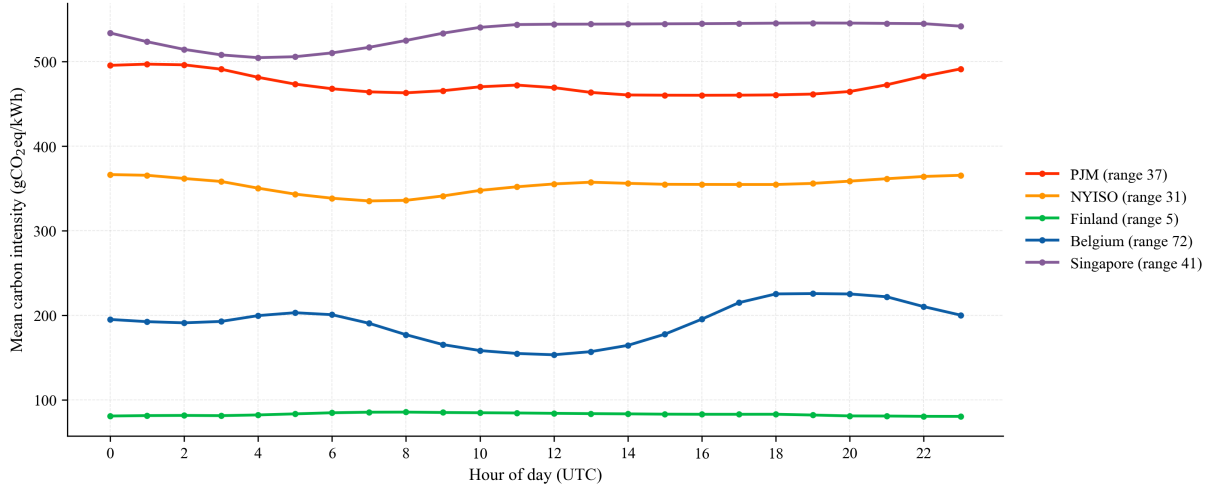


Figure A.1: Average carbon intensity by hour of day (UTC) per region. The intra-day range (shown in the legend) is largest for Belgium, driven by midday solar, and smallest for Finland, whose carbon intensity is nearly flat across the day despite being the most variable region overall on a multi-day basis.

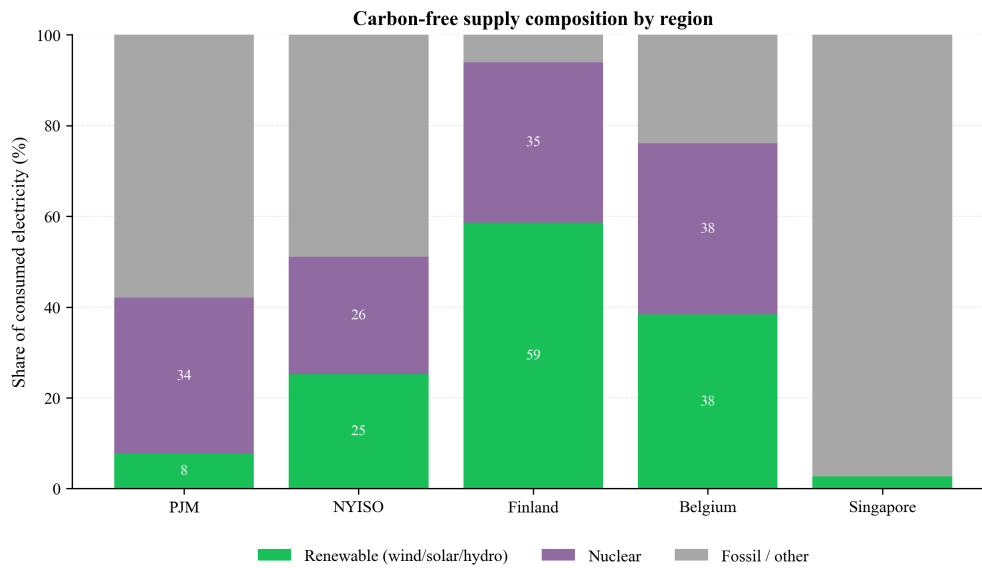


Figure A.2: Mean composition of carbon-free supply by region, split into renewable (wind, solar, hydro) and nuclear shares. PJM and NYISO reach much of their carbon-free fraction through nuclear rather than renewables, which is why a renewables-only signal would misrepresent their cleanliness and the carbon-free energy fraction is used instead.

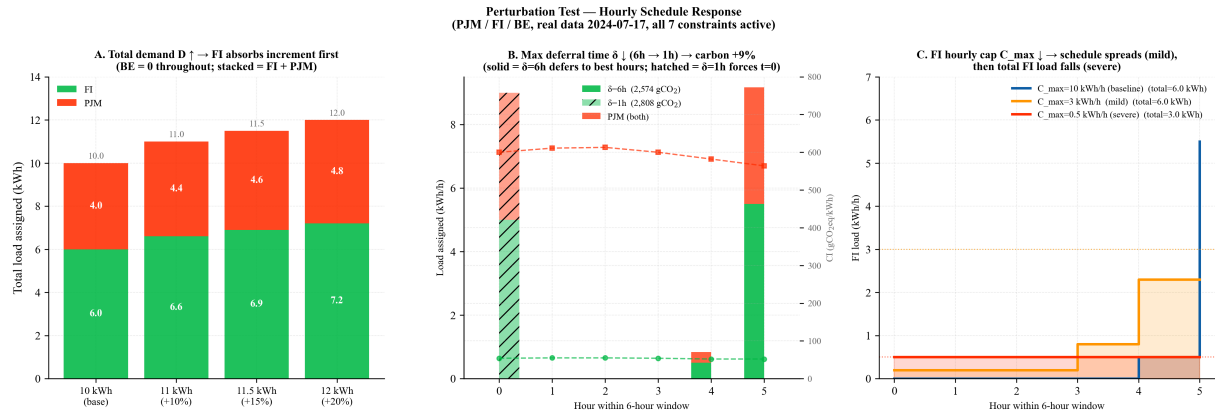


Figure A.3: Hourly schedules for three of the perturbation scenarios in Table 3 (the CI shock is reported in the table but not plotted). Demand increments flow to the cleanest region (left), a shorter deferral window forces load out of the clean hours and raises carbon (centre), and a tighter hourly cap first spreads Finland’s load and then reduces it (right).

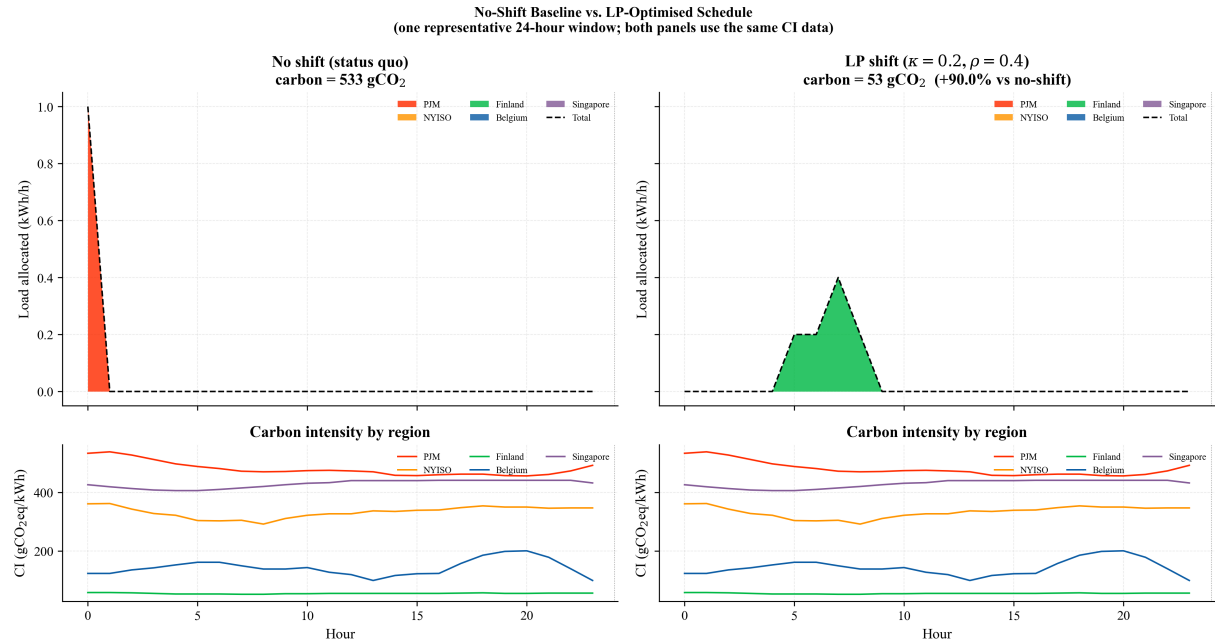


Figure A.4: No-shift baseline versus the LP-optimized schedule for one representative window, with the carbon-intensity signal below. The LP routes the batch to Finland and spreads it across hours to satisfy the ramp limit, cutting window carbon from 533 to 53 gCO₂.

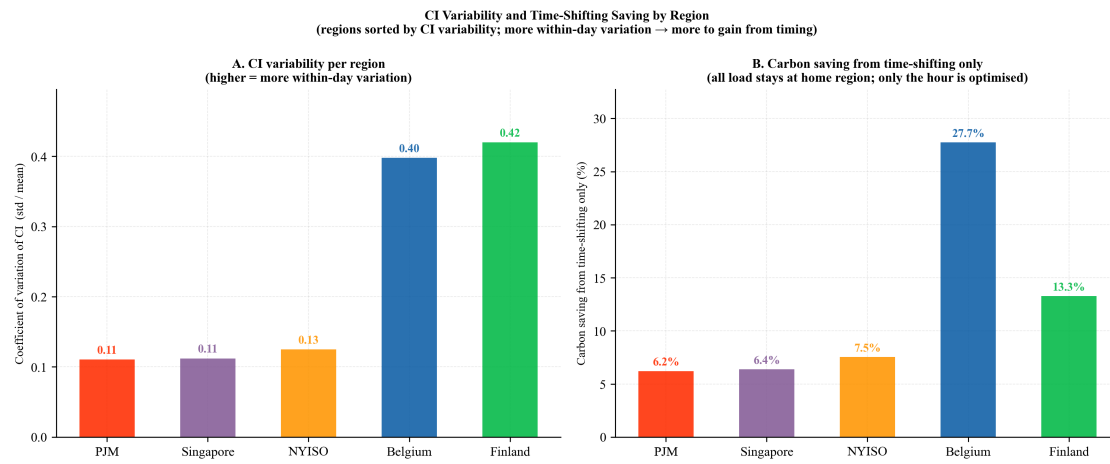


Figure A.5: Within-region carbon saving from timing alone against the coefficient of variation of carbon intensity, one point per region. Timing helps materially only in the high-variability grids (Belgium, Finland) and barely at all in the flat ones (PJM, NYISO, Singapore).